VERONICA SCOZZI

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Citizenship: Italian | Current Postion: Scientific Intern in Vogels Lab at ISTA, Austria
github profile | Online Portfolio

EDUCATION

Bachelor of Science in Engineering (BSc) in Informatics Sept 2023 – Present
IMC University of Applied Sciences, Austria *Expected Graduation: July 2027*

- **Current Cumulative Average:** 90% (≈US A-)
- **Last Academic Year GPA:** 1.36 / 5.00 *(Austrian Scale: 1 is best, 5 is worst ≈ US 3.8/3.9)*
- **Honors:** Recipient of an excellence scholarship (2025) for outstanding academic achievements.

CONFERENCES & PREPRINTS

- **Scozzi, V.**, Condruz, N., Vogels, T. P. (2026). *Robustness of Spectral and Dynamical Properties in Cortical Network Models Operating Near Instability*. Presented a Poster at the **Bernstein Conference 2026**, Frankfurt am Main, Germany. (September 2026). abstract
- **Scozzi, V.**, Di Troia, F. (release on bioRxiv expected by late 2026/ early 2027). *Regularization Improves the performance of Data-efficient Linear Ensemble Encoding Model on Voxel-wise Prediction Task*. **Manuscript in preparation**.
- **Scozzi, V.**, Condruz, N., Vogels, T. P. (release on bioRxiv expected by early 2027). *Robustness of Spectral and Dynamical Properties in Cortical Network Models Operating Near Instability*. **Manuscript in preparation**.

RESEARCH EXPERIENCE

Full-time Scientific Intern | Vogels Lab
Institute of Science and Technology Austria (ISTA) Mid-Jan 2026 – Mid-Jan 2027 (Ongoing)

- Initiated a 6-month exploratory internship to bridge an Informatics background into Theoretical Neuroscience, learned about biological and mathematical foundations of the lab's science through the replication and expansion of network benchmarks (including FORCE learning, Stability Optimized Circuit models, and continuous attractor models).
- Received a 6-month contract extension to transition into an independent research track focused on the study of cortical network properties under the supervision of Prof. Tim Vogels.
- Implemented a network model showcasing rich dynamical range comparable to those observed in motor cortex.
- Perturbed the structure with different perturbation types and analyzed their impact and accumulation effects.
- Presented original findings in a poster session at the international **Bernstein Conference 2026** (September 28th, 2026 - October 1st, 2026).
- Currently investigating different types of compensatory mechanisms to track the recovery of spectral and complex dynamical properties.
- Currently drafting a first-author manuscript for journal submission by the end of the internship.

Undergraduate Student Researcher (Part-time)
Remote Research Collaboration | Mentor: Prof. Fabio Di Troia, San Jose State University March 2025 – Ongoing

- Conceptualized and built a data-efficient transfer learning framework to construct neural encoding models using the **Natural Scenes Dataset (NSD)**.
- Replicated the methodology of *Gu et al. (2022)*, identified a limitation in voxel-wise prediction accuracy, performed systematic error analysis and found a working solution.
- Engineered data pipelines to construct resting-state functional connectivity fingerprints and analyzed their structure.
- Currently drafting a first-author manuscript presenting the adjusted encoding architecture, the robustness of the approach, the results achieved, and the control analysis performed.
- We will further expand the work and the framework by adding the subject's connectivity fingerprint as a structural constraint to condition the model's predictions. Improvements will be published in a subsequent paper.

Please refer to the online portfolio for extended documentation, plots, figures and code-bases for all research and academic projects.

RELEVANT TECHNICAL SKILLS

- **Computational Neuroscience & ML:** Firing-rate models, Spiking Neural Networks (SNN) such as LIF models, fMRI & EEG Signal Processing, PyTorch, NEST, Brian2, Neural Decoding, Spectral Analysis (PSD).
- **Languages & Core CS:** Python, R, C, JavaScript, SQL, Linux Bash, Systems Programming.
- **Mathematics & Theory:** Linear Algebra, Multivariable Calculus, Probability & Statistics, Dynamical Systems, Information Theory.
- **Tools & Frameworks:** Git/GitHub, LaTeX, Docker, Neo4j, Django.

SELECTED ACADEMIC PROJECTS

Automated Sleep Stages Classification from Raw EEG Data 2025

- Engineered a bilingual pipeline (Python for FIR filtering/PSD extraction, R for modeling) implementing XGBoost, SVM, Random Forrest, KNN and FNN and more architectures on a subset of the public BOAS dataset. Validated the work using Leave-One-Subject-Out (LOSO) Cross-Validation, Pooled 10-fold Cross-validation and true Within-Subject 10-fold Cross-Validation. Code available on GitHub.

Single neuron and Networks Modeling Projects 2026

- Implemented 5 projects of various complexities, scales and approaches. From a single detailed LIF neuron to a complex 200-neuron Hennequin's 2014 SOC network, including continuos attractor networks such as Ben-Yishai's 1995 ring attractor model and learning algorithms such as FORCE learning implemented on various architectures of firing-rate models. Detailed information can be found on my portfolio.

The NSD Mapping Project: A Systematic Methodological Compendium of Natural Scenes Dataset Studies. 2025

- Authored a 33-page open-source methodological compendium systematically indexing and analyzing the computational architectures, encoding/decoding models, and mathematical frameworks from 20 7T fMRI Natural Scenes Dataset publications.

HONORS & AWARDS

- **Academic Merit Scholarship** | IMC University of Applied Sciences 2025
  This scholarship is awarded every year by IMC Krems to the top students from each degree program for "outstanding academic achievements". For my program specifically, only about 10 scholarships are available each year.
- **Neuroscience Olympics (International Brain Bee)** | 2nd Place Regional Phase (Sardinia, Italy) 2023

LANGUAGES

- **Italian:** Native proficiency.
- **English:** Proficient User (C2) – *IELTS Academic: 8.5* (2026).